

📰 Fake News Detector using Machine Learning and Streamlit

This is a simple and interactive web application that uses a Logistic Regression model to detect whether a news article is ****real or fake****. Built using Python, scikit-learn, and Streamlit.

🚀 Features

- Predicts whether a news article is ****Fake**** or ****Real****
- Machine learning model trained using TF-IDF and Logistic Regression
- Clean UI with text area input and instant prediction
- Ready to deploy on Streamlit Cloud

🛠️ Tech Stack

- Python
- Pandas, scikit-learn, joblib

- Streamlit (for the web UI)
- Logistic Regression Classifier
- TF-IDF Vectorizer

📁 Dataset

Used two datasets:

- `Fake.csv` – Collection of fake news articles
- `True.csv` – Collection of real news articles

These were merged and preprocessed to train the model.

☐ Model Training (Summary)

- Dropped irrelevant columns like `title`, `subject`, `date`
- Cleaned text (lowercasing, removing punctuation, digits, links)
- Used `TfidfVectorizer` for feature extraction
- Trained using `LogisticRegression`

- Saved using `joblib` for deployment

📖 How to Run Locally

1. Create virtual environment (optional but recommended)

```
python -m venv venv    (for windows)
```

```
venv\Scripts\activate
```

2. Install dependencies

```
pip install -r requirements.txt
```

3. Run the app

```
streamlit run app.py
```